Tectonic Rotations Suggested by Paleomagnetic Results from Northern Colombia, South America[1]

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Nine sites in two Cretaceous units, the La Teta lava and the Maruayan diorite (approximately 12°N, 72°W), from the Guajira peninsula of Colombia give a virtual geomagnetic pole in central equatorial Africa. The Cretaceous pole for eight sites in La Teta lava is located at 10°N, 21°E ($\delta p = 10°$, $\delta m = 19°$). This pole is about 90° from Cretaceous poles determined by other workers for North and South America. Our results, combined with the results of other workers from circum-Caribbean sites, are interpreted in terms of large tectonic rotations of Caribbean sites during and after Cretaceous times. A paleotectonic reconstruction of the Americas was made by superimposing Cretaceous virtual paleomagnetic poles. It is concluded that the Guajira and Greater Antilles sites occupied latitudes about 10° south of their present latitudes, possibly off northwestern South America, in Cretaceous times. Eleven other sites from two diorite bodies in the Guajira peninsula gave inconclusive results.

The Guajira peninsula is an arid region in northernmost South America, centrally located on the south shores of the Caribbean Sea (Figure 1). Pre-Tertiary rocks here are exposed in isolated topographic and structural highs surrounded by lowlands of flat-lying Tertiary marine strata.

The pre-Tertiary rocks of the peninsula form parallel structural zones of ENE trend [*MacDonald et al.*, 1968]. One hundred and twenty-three oriented cores were taken at 21 sites in the vicinity of three diorite bodies intrusive into the pre-Tertiary terrains. Six units were sampled. Three are diorite intrusions, of late Paleozoic-Early Triassic(?) age (Siapana diorite), Cretaceous age (Maruayan diorite), and Eocene age (Parashi diorite). Dikes in two of the diorites were cored (Siapana and Parashi dikes). Extensive Cretaceous siliceous flows (La Teta lava) intruded by the Maruayan diorite were also sampled.

Five to eight samples were taken at each site by using a portable gasoline-powered drill that yields cores 2.45 cm in diameter. The cores were oriented by using a magnetic compass. A sun compass was used to check the orientation at 13 of the 21 sites. At most localities deviations of the corrected magnetic azimuth from the solar azimuth were <2°, except for observations near midday. Therefore the magnetic orientations, which have been used in the calculations, are generally believed to be accurate within about ±2°.

Siapana diorite and dikes. Of five sites in the Siapana diorite, three are diorite, and two are andesite dikes cutting the diorite. The Siapana diorite, actually granodioritic in composition, is believed to be approximately Permo-Triassic in age. A K-Ar age value of 195 ± 8 m.y. has been obtained from muscovite in a pegmatite close to the intrusion. This age value and those age values from the La Teta lava are thought to be minimums because of argon loss during Late Cretaceous to early Tertiary tectonic activity [*MacDonald*, 1968]. No ages have been determined for the Siapana dikes.

La Teta lava and Maruayan diorite. Eight sites were cored in the La Teta lava, and one

[1] Lamont-Doherty Geological Observatory contribution 1849.

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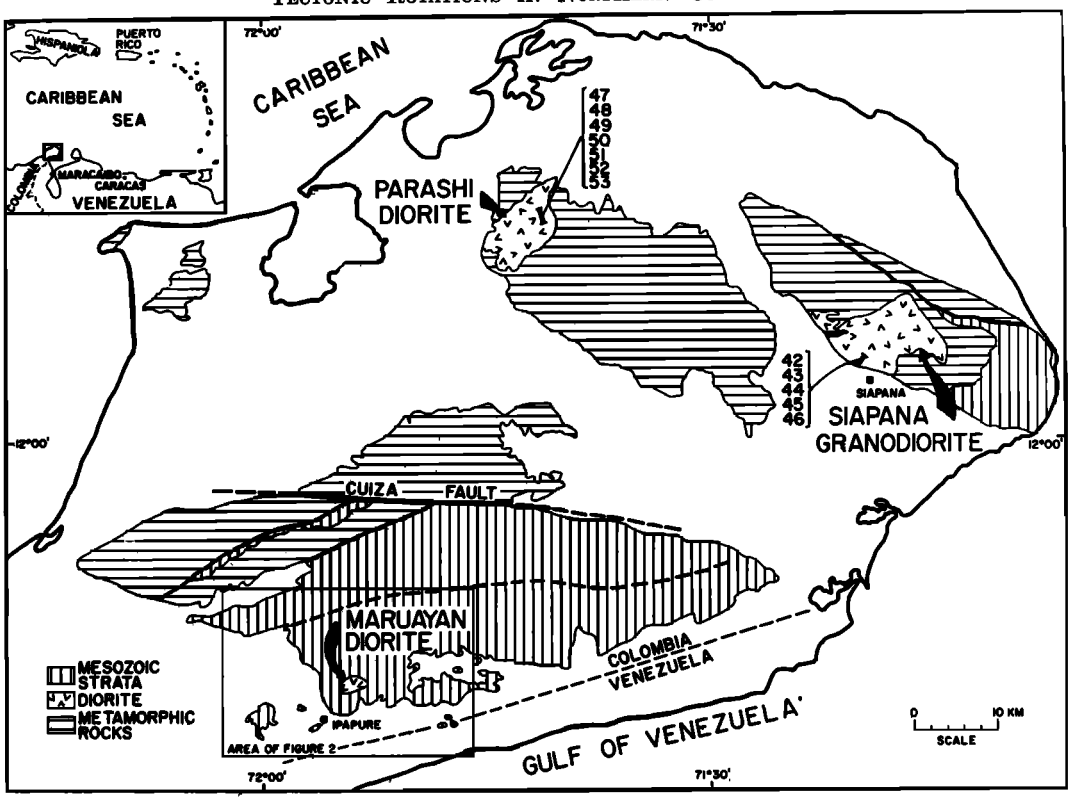


Fig. 1. Geologic sketch map of Guajira peninsula, Colombia and Venezuela, with paleomagnetic sampling sites.

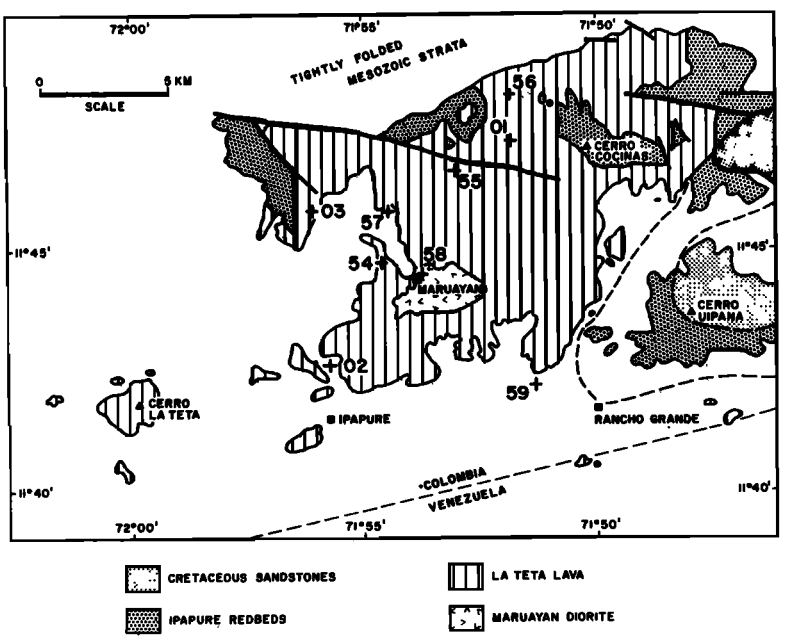


Fig. 2. Geologic map of Cretaceous formations with locations of sample sites in the La Teta lava and the Maruayan diorite. The area is outlined in Figure 1.

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TABLE 1. Site Statistics for Siapana Diorite and Dikes

Site	Location	Cores	No. of Specimens	Natural Remanent Magnetism				Alternating Field Demagnetized				
				Dm, deg	*Im*, deg	κ	α_{95}, deg	Field, oe	*Dm*, deg	*Im*, deg	κ	α_{95}, deg
				Siapana Diorite								
42	12.08°N, 71.32°W	6	9	315.6	45.8	1.8	57.6	100	337.5	61.8	2.8	37.9
45	12.08°N, 71.32°W	6	9	233.6	53.4	1.2		50	262.5	53.0	2.0	51.1
46	12.08°N, 71.32°W	6	8	359.9	43.6	1.4		75	251.2	69.2	2.1	53.0
				Siapana Dikes								
43	12.08°N, 71.32°W	7	9	300.8	-8.3	2.7	38.7	200	264.2	-38.2	4.2	30.8
44	12.08°N, 71.32°W	6	8	289.1	28.3	1.5	74.3	100	245.1	-62.6	3.1	37.6

Dm, declination; *Im*, inclination; κ, Fisher's precision parameter; α_{95}, radius of 95% confidence circle.

TABLE 2. Site Statistics for La Teta Lava and Maruayan Diorite

Site	Location	Cores	No. of Specimens	Natural Remanent Magnetism				Alternating Field Demagnetized				
				Dm, deg	*Im*, deg	κ	α_{95}, deg	Field, oe	*Dm*, deg	*Im*, deg	κ	α_{95}, deg
				La Teta Lava								
1	11.78°N, 71.86°W	5	6	290.0	-3.1	4.7	39.6	100	273.6	-5.2	70.2	8.1
2	11.71°N, 71.93°W	4	4	87.8	-11.3	175.6	7.0	100	89.6	-7.8	169.1	7.1
3	11.76°N, 71.93°W	5	5	238.9	26.7	1.9	79.2	100	247.9	27.9	598.7	3.1
54	11.75°N, 71.91°W	6	6	243.3	29.4	14.3	18.3	200	232.4	10.1	15.9	17.3
55	11.78°N, 71.91°W	6	6	295.9	-5.8	81.6	7.5	300	291.5	-15.7	82.3	7.4
56	11.80°N. 71.87°W	5	5	247.9	-18.5	1.9	68.1	300	268.1	-2.0	765.4	2.8
57	11.77°N, 71.91°W	6	6	245.7	-24.8	43.4	10.3	300	249.4	-21.4	352.8	3.6
59	11.70°N, 71.86°W	6	6	248.3	21.6	2.4	55.7	300	242.4	-2.8	16.5	19.4
				Maruayan Diorite								
58	11.74°N, 71.90°W	6	6	350.8	32.5	5.0	33.4	200	266.4	-31.1	6.6	28.2

Dm, declination; *Im*, inclination; κ, Fisher's precision parameter; α_{95}, radius of 95% confidence circle.

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was cored in the Maruayan diorite (Figure 2). The La Teta lava forms a thick flat-lying sequence of rhyodacitic flows intruded by the small Maruayan stock. The La Teta lava lies below a thick redbed sequence (Ipapure redbeds, Figure 2), which in turn lies below a thick sequence of Cretaceous sandstones. According to *Rollins* [1965], the sandstone contains Berriasian to Valanginian (earliest Cretaceous) fossils. K-Ar age values of 120 and 95 m.y. were obtained from the La Teta lava. Stratigraphic relationships indicate these Cretaceous ages to be minimums. The Maruayan diorite also gave a Cretaceous whole-rock K-Ar age value of 120 ± 4 m.y.

Parashi diorite and dikes. Of seven sites in the Parashi diorite, two were cored in diorite, and five were cored in dikes cutting the diorite. The dikes are remarkably abundant and subparallel in a N-S direction. They are mainly of hornblende dacite porphyry [*Lockwood*, 1965]. The age of the Parashi dikes is thought to be close to that of the Eocene diorite.

LABORATORY PROCEDURES

The direction and the intensity of all specimens were measured on a 5-Hz spinner magnetometer [*Foster*, 1966] or a commercially available 105-Hz spinner magnetometer. Alternating-field demagnetization was carried out in steps following the procedures described by *McElhinny and Gough* [1963].

Siapana diorite and dikes. The specimens from the Siapana sites possess variable intensities of magnetization, both within and between sites, ranging from 10^{-5} emu/cm^3 for the dikes to 10^{-6} emu/cm^3 for the diorite.

The initial directions of magnetization for both diorite and dikes are widely dispersed (Table 1). Alternating-field demagnetization improves the grouping only marginally, although the directions from the dikes indicate a rough grouping towards the WSW with negative inclinations. Attempts to improve the groupings by using higher fields were unsuccessful. The af demagnetization curves (Figure 3) indicate a large unstable component that is removed in very low fields. The median destructive field (mdf) is <25 oe. The data are insufficient to draw any paleomagnetically meaningful interpretation, and no pole position has been calculated from them. Although declinations vary widely after ac demagnetization, the diorite specimens have positive inclinations, and the dikes have negative inclinations, three out of 43 specimens deviating from this pattern. The dikes might yield significant results if more sites could be obtained.

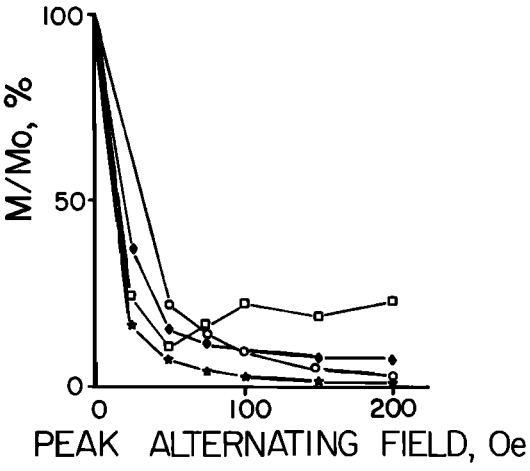


Fig. 3. Alternating-field partial-demagnetization curves of specimens from the Siapana dikes and diorite and from the Parashi dikes and diorite. The per cent intensity of magnetization of the sample relative to its NRM is plotted as the ordinate. The ac demagnetizing field strength in oersteds is plotted as the abscissa. Squares indicate specimen 44, Siapana dike; diamonds, specimen 42, Siapana diorite; stars, specimen 47, Parashi dike; circles, specimen 49, Parashi diorite.

La Teta lava and Maruayan diorite. The NRM directions of the La Teta lava are initially well grouped (Table 2), with intensities of magnetization ranging from 10^{-3} to 10^{-5} emu/cm^3 with southwestward declinations and shallow inclinations, the exception being site 2, which is magnetized toward the east, again with shallow inclination. These rocks possess a very stable magnetization, as can be seen in Figure 4, which shows that quite high fields are necessary to decrease the intensity of magnetization to 50% of its initial value. In the specimen figured, this value is 350 oe. This value represents very stable behavior, especially when it is compared with values from rocks such as the Siapana diorite. Stability is also indicated by the reversed direction at site 2, which is approximately 180° reversed with respect to the other sites. Alternating-field demagnetization serves to decrease the within-site scatter. The mean directions of magnetization with circles of confidence are shown in Figure 5.

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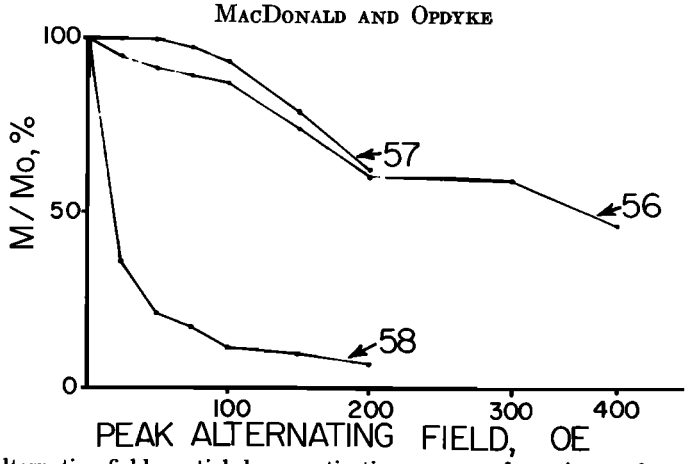


Fig. 4. Alternating-field partial-demagnetization curves of specimens from the La Teta lava and the Maruayan diorite. The ordinate and the abscissa are as in Figure 3. Specimens 56 and 57 are from the La Teta lavas; specimen 58 is from the Maruayan diorite.

Fisher's [1953] statistics were calculated from the site mean directions after the directions at site 2 were reversed and are $Dm = 259.4°$, $Im = -0.2°$, $\kappa = 10.9$, $\alpha_{95} = 17.5°$, $R = 7.4$, and $N = 8$.

The single site from the Maruayan diorite has initial directions of NRM directed to the north with positive inclinations almost in the direction of the earth's present field. Figure 4 shows an af demagnetization curve from this site. The intensity of magnetization drops sharply in low fields, and thus a large magnetically 'soft' component is shown to be present. After af demagnetization in fields of 200 oe the direction of magnetization has moved close to the directions of magnetization of the lavas, and it is thus suggested that the diorite was magnetized at approximately the same time as the lavas. Because the K-Ar whole-rock age of the diorite is close to that of the lava, this assumption would appear to be valid.

Parashi diorite and dikes. With the exception of site 52, the directions of the nrm of both the dikes and the diorite are directed toward the north with positive inclinations (Table 3). The intensities fall in the range 10^{-3}–10^{-4} emu/cm^3. The af demagnetization causes a sharp drop in intensity in low fields (Figure 3). The mdf is <50 oe. The af demagnetization in low fields led to reasonable

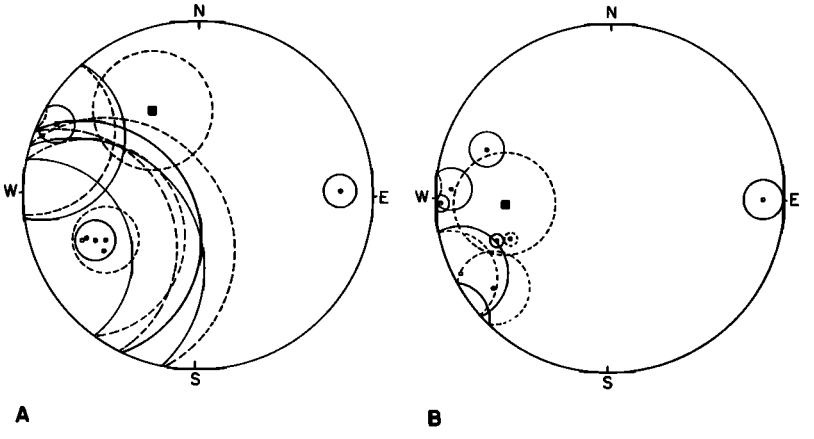


Fig. 5. Site mean directions of magnetization for the La Teta lava (small circles) and the Maruayan diorite (squares). Closed circles indicate lower-hemisphere projections, and open circles indicate upper-hemisphere projections. Equal-angle projection with the local horizons prime. *a*) NRM. *b*) ac demagnetized. The vectors plotted are from Table 2. Circles of 95% confidence are drawn around each site mean direction.

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within-site groupings in many cases. However, the between-site scatter was very large. Some improvement in the between-site groupings was made by af partial demagnetization in fields of 500 oe. However, even with the improvement, the declinations for the site range over more than 180° of arc. The inclinations are positive and range between 28° and 57°, so that no true reversal is present. The large scatter between site mean directions is difficult to account for. One possibility is that the dikes possess a secondary component of magnetization that we have been unable to remove. It is also possible that the dikes were intruded over a long time, during which the region was subjected to tectonic rotation. No pole position has been calculated from these widely dispersed data.

DISCUSSION AND INTERPRETATION

A reliable pole position has been determined for the La Teta lava only. The magnetization of the lava is stable, and there is good grouping of the directions of magnetization, both within and between the sites.

Attempts to improve further the clustering of the La Teta directions of magnetization by making bedding corrections were inconclusive. Bedding is generally obscure in these thick and massive flows. A plot of nine poles to flow layering (Figure 6*a*) shows a slightly greater variation in orientation than a similar plot of 62 poles to bedding in clastic rocks overlying the flows (Figure 6*b*). This variation is thought to be due to the large variation of initial dip inherent in the flow layering. Therefore it is not too surprising that corrections applied for flow layering, which is assumed to be initially horizontal, resulted in only slight improvement in the grouping of the site mean directions. For the four sampling sites (1, 2, 3, and 56) where flow layering measurements were possible, Fisher statistics of the mean directions of magnetization corrected for tilt are $Dm = 271.3°$, $Im = -0.3°$, $\kappa = 10.0$, $\alpha_{95} = 30.6°$, $R = 3.7$, and $N = 4$. In calculating the over-all mean af demagnetized direction of magnetization for the eight lava sites, no tilting corrections were applied.

The pole location for the La Teta lava (Table 4) is based on the assumption that the lava unit is horizontal; this assumption seems to be reasonable on the basis of the

TABLE 3. Site Statistics for Parashi Diorite and Dikes

Site	Location		Cores	No. of Specimens	Natural Remanent Magnetism				Alternating Field Demagnetized				
					Dm, deg	*Im*, deg	κ	α_{95}, deg	Field, oe	*Dm*, deg	*Im*, deg	κ	α_{95}, deg
					Parashi Diorite								
49	12.25°N,	71.68°W	6	7	001.5	27.6	6.2	26.3	500	329.7	44.4	5.3	28.7
50	12.24°N,	71.68°W	6	7	241.0	28.3	14.5	16.4	500	239.3	28.1	23.6	12.6
					Parashi Dikes								
47	12.25°N,	71.68°W	6	11	323.8	23.8	3.2	30.6	500	297.6	40.6	12.3	13.5
48	12.25°N,	71.68°W	6	9	25.6	36.6	4.7	26.5	500	264.3	44.3	7.2	20.5
51	12.24°N,	71.68°W	6	10	46.6	33.6	52.5	6.7	500	56.8	33.9	27.0	9.4
52	12.25°N,	71.68°W	6	8	103.7	13.7	9.9	18.5	500	91.1	57.2	12.0	16.6
53	12.25°N,	71.68°W	6	6	18.0	28.6	20.7	15.1	500	345.4	37.3	55.2	9.0

Dm, declination; *Im*, inclination; κ, Fisher's precision parameter; α_{95}, radius of 95% confidence circle.

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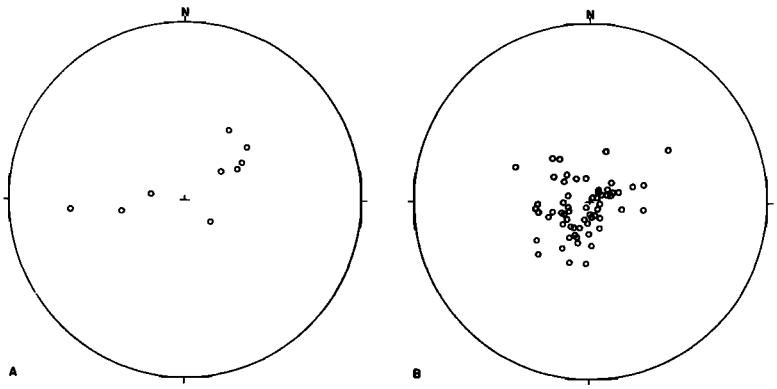


Fig. 6. A comparison of orientations of *a*) flow layering in the La Teta lava with *b*) bedding in overlying redbeds and sandstones. The scatter in the flow layering suggests that 'initial dip' in flow layering may be steep. The clustering of poles to bedding close to the center of the projection shows that the bedding of the sampling area is approximately horizontal. Equal-angle projection with the local horizons prime.

good clustering of the directions of magnetization and the horizontality of the overlying beds. The bedding measurements (Figure 6*b*) indicate that the beds immediately above the flows are little deformed. These bedding data were taken from *Renz* [1960], *Rollins* [1960], and from our own observations in the area shown in Figure 2. Most of the measurements were made in the redbeds overlying the lava flows. The mean dip of these beds is very close to horizontal with a slight possible regional dip toward the north.

The mean inclination for 45 specimens of the La Teta lava is about 1° ± 8°. This inclination is about 12° less than that of the theoretical dipole field for the present latitude (inclination 23° at latitude 11.8°) and suggests that in Cretaceous times the La Teta lava lay close to the equator. The present field in this region is inclined 43° [*U.S. Naval Oceanographic Office,* 1966]. The declinations for the La Teta lava lie at an angle of almost 90° to the present field and are at large angles to the Cretaceous declinations for sites in both North and South America, as is discussed below.

The Maruayan vectors are closely parallel to those of the La Teta lava. Too few cores were taken in the Maruayan diorite to merit determining a pole position for that body. Inclusion of the Maruayan data does not significantly change the La Teta pole position (Table 4).

SIGNIFICANCE IN RELATION TO OTHER RESULTS

Cretaceous pole positions obtained from sites in the Caribbean area and in North and South America are shown in Figure 7 and tabulated

TABLE 4. Rock Unit Statistics and Corresponding Paleomagnetic Pole Locations

Unit	No. of Sites	Mean of Site Means				Pole Positions			
		D_m, deg	I_m, deg	κ	α_{95}, deg	Location	p, deg	δp, deg	δm, deg
La Teta lava (af demagnetized)	8	259.4	0.6	9.3	19.1	10.3°N, 20.6°E	89.7	9.6	19.1
La Teta lava and Maruayan diorite (af demagnetized)	9	260.1	4.2	9.1	18.0	9.2°N, 22.2°E	87.9	9.0	18.0

D_m, declination; I_m, inclination; κ, Fisher's precision parameter; α_{95}, radius of 95% confidence circle; p, colatitude; δp, radius of confidence oval measured in the direction from pole toward site; δm, radius of confidence oval measured perpendicular to δp.

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in Table 5. *Larochelle* [1968] has shown that there is good agreement between the Cretaceous poles for North American rocks. Unfortunately, data for South America are more scarce. *Creer's* [1970] preliminary Cretaceous poles for Peru, Colombia, and Venezuela lie within about 20° of *Creer's* [1962] pole for the Early Cretaceous Serra Geral basalts of Brazil.

Relative to the well-controlled Cretaceous poles for the American continents, such as *Larochelle's* [1968] pole for North America and *Creer's* [1962] for South America, most of the Caribbean Cretaceous poles are displaced. They are also displaced relative to one another, as is clear from Figure 7. *Watkins and Cambray* [1970] reported two Cretaceous poles from dikes in Jamaica. One lies close to the Cretaceous pole of North America, and the other is close to the Cretaceous Guajira pole of this study. S. A. Vincenz (personal communication, 1971) and *Guja and Vincenz* [1971] have reported Cretaceous poles from Jamaica that lie in middle- and low-latitude positions. *Fink and Harrison* [1971] reported a Cretaceous pole from Puerto Rico that is similarly disposed.

The interpretation of the paleomagnetic data from the Caribbean is subject to great uncertainties as long as data remain so few. *MacDonald and Opdyke* [1970] suggested tectonic rotation as the explanation for the equatorial location of the Guajira Cretaceous pole. *Watkins and Cambray* [1970] have interpreted the equatorial position of one of the Cretaceous poles of Jamaica as being due to a temporary excursion of the dipole field.

We favor the tectonic rotation hypothesis for two reasons. First, the magnetization of the Cretaceous sites of the Guajira must represent a considerable span of time, since similar directions of magnetization are found in both the Maruayan diorite stock and the La Teta flows. The lavas themselves were extruded over a period of time long enough to include at least one reversal of the field. These observations imply a stable position for the dipole field in equatorial latitudes, a condition which on theoretical grounds seems unlikely. Second, from Figure 7 it can be appreciated that the various Cretaceous poles from the Caribbean region outline an arc about the Caribbean. This configuration is to be expected, if tectonic rotation is the explanation for the anomalous positions of the Caribbean poles. If the explanation lies in an anomalous temporary geomagnetic field, the resulting dispersion of Cretaceous poles should be equally evident in data from tectonically stable areas of North America. Such dispersion is not apparent.



Fig. 7. Caribbean Cretaceous paleomagnetic pole positions compared with those of North and South America. Paleomagnetic poles are shown by closed circles, and the corresponding sites are shown by plus signs. Numbers refer to sites in Table 5. Paleomagnetic poles of circum-Caribbean sites (7–13) are greatly displaced from those for North and South America (1 and 2). Circum-Caribbean sites give paleomagnetic poles forming a rough arc around the Caribbean at a distance of 80°–90°. This configuration suggests Cretaceous and younger tectonic rotations of the circum-Caribbean sites and Cretaceous latitudes between 0° and 10° for these sites. Equal-area projection.

RECONSTRUCTION BASED ON PALEOLATITUDES

The evolution of the Caribbean plate in terms of ocean-floor spreading is still unknown. An analysis of some of the dilemmas posed by the Caribbean plate is given by *Meyerhoff and Meyerhoff* [1972]. A recent reconstruction [*Smith et al.*, 1972] suggests that the Caribbean may once have been a southwestward continuation of an early North Atlantic ocean. An interpretation of the available but scanty

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TABLE 5. Caribbean Cretaceous Pole Locations Relative to Those for the American Continents

Site Location*	Present Colatitude, deg	Pole Location			Cretaceous Paleolatitude,¶ deg	Reference
		Location	δp,† deg	δm,§ deg		
			North America			
1 45°N, 73°W		69°S, 8°E				*Larochelle* [1968]
			South America			
Brazil						
2 24°S, 54°W		78°S, 54°E	6	6		*Creer* [1962]
Peru						
3 12°S, 77°W		63°S, 30°E				*Creer* [1970]
Colombia						
4 05°N, 73°W		80°S, 72°W				*Creer* [1970]
Venezuela						
5 09°N, 68°W		68°N, 53°E				*Creer* [1970]
6 09°N, 68°W		81°S, 120°W				*Creer* [1970]
			Caribbean			
Jamaica						
7 18°N, 77°W	72	70°S, 12°W	10	21	9	*Watkins and Cambray* [1970]
8 18°N, 77°W	72	12°S, 09°E	13	27	0	*Watkins and Cambray* [1970]
9 18°N, 77°W	72	43°S, 36°E			29	S. A. Vincenz (personal communication, 1971)‖
10 18°N, 77°W	72	12°S, 04°E			6	*Guja and Vincenz* [1971]
11 18°N, 77°W	72	01°S, 21°E			8	*Guja and Vincenz* [1971]
Guajira, Colombia						
12 12°N, 72°W	78	10°N, 20°E	10	19	0	This study
Puerto Rico						
13 18°N, 67°W	72	35°S, 31°E			15	*Fink and Harrison* [1971]

Poles have been transposed where necessary to the hemisphere west of longitude 60°E to be plotted in Figure 10.
*Site locations are approximate.
†Radius of confidence oval measured in the direction from pole toward site.
§Radius of confidence oval measured perpendicular to δp.
¶Cretaceous paleolatitudes are relative to the closest pole and have been calculated from data in this table.
‖*Vincenz* [1971] reports this pole as 46°S, 36°E, α_{95} = 12°.

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Cretaceous paleomagnetic data from Caribbean sites is given below.

Most of the paleomagnetic information from the Greater Antilles Cretaceous is from Jamaica. Cretaceous paleolatitudes for Jamaica (Table 5) are mainly $<10°$. The present latitude of Jamaica is 18°. The paleomagnetic pole of North America shows little motion in the Cretaceous period [*Larochelle*, 1968]. Relative to that pole the Cretaceous paleolatitude of a fixed Jamaica should be 14° ± 2.8°. Relative to the best available pole for the Cretaceous of South America [*Creer*, 1962] the Cretaceous paleolatitude of a fixed Jamaica should be 25° ± 6°. At first sight it appears that Jamaica has moved northward relative to the equator and has remained fixed relative to neither North or South America since Cretaceous time. It should be noted, however, that direct comparisons between Jamaica and South America are uncertain because the published Cretaceous data are not exactly coeval. The Serra Geral pole of Brazil (1962) represents Early Cretaceous time, whereas the Jamaican data represent Late Cretaceous time.

Cretaceous or post-Cretaceous latitudinal shifts are also indicated in data from other Caribbean sites (Table 5). The Cretaceous paleolatitude for the La Teta lava is 0°; the present latitude is 12°. The paleolatitude from a single pole for Puerto Rico is estimated to be 15°; the present latitude is 18°.

Most of the available paleomagnetic data for the Cretaceous circum-Caribbean sites suggest Cretaceous paleolatitudes 5°–10° less than the present latitudes and Cretaceous paleomagnetic poles arranged approximately in an arc around the Caribbean. These data suggest motion of the Caribbean sites relative to both North and South America since Cretaceous time. A post-Cretaceous shift of circum-Caribbean sites away from more southerly (equatorial) Cretaceous latitudes is also indicated.

A reconstruction based on the Cretaceous paleolatitudes is obtained by superimposing the Cretaceous paleomagnetic poles (Figure 8). The relative latitudes of North America, South America, and parts of the Caribbean region can be represented thus for Cretaceous time. There is no control of longitude in this reconstruction, except that overlap of land areas is not permissible. The Cretaceous pole of *Larochelle* [1968] is used for North America, and that of *Creer* [1962] is used for South America. The present latitude and longitude grids are shown on those continents. For the Caribbean sites the following paleolatitudes have been used: Jamaica, 5°; Guajira, 0°; and Puerto Rico, 15°.

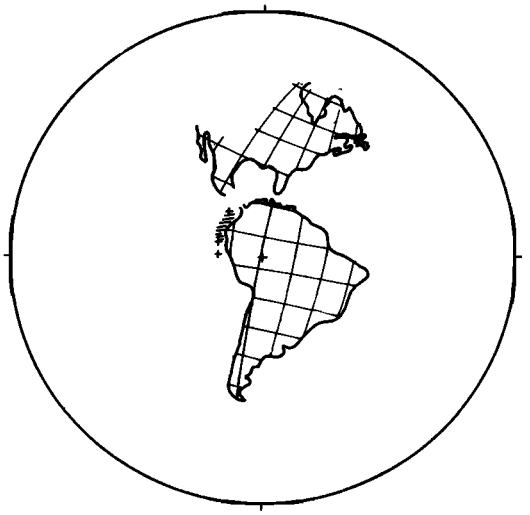


Fig. 8. A paleotectonic reconstruction of the Americas based on Cretaceous paleolatitudes (Table 5). The poles of *Larochelle* [1968] and of *Creer* [1962] for North and South America, respectively, have been superimposed on those for Guajira, Jamaica, and Puerto Rico (see text). The slanted shading indicates the latitude range thought to have been occupied by the Greater Antilles in Cretaceous times. The three plus signs off western South America represent, from south to north, the latitudinal positions of the Guajira peninsula, Jamaica, and Puerto Rico in Cretaceous times. In this reconstruction there is no control on relative longitudes. Equal-area projection with center marked by large plus sign.

It is suggested that the Caribbean plate, together with some of the present surrounding land masses, has moved northward and eastward since Cretaceous time relative to northwest South America. This reconstruction is similar to that of *Bullard et al.* [1965], North America being placed in a slightly more westerly position relative to South America. An implication of this reconstruction is that microcontinental plates of the Yucatan and Nicaraguan regions may also have derived from positions further south. Possibly they were torn from the northwest part of South America.

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Previous reconstructions have shown them being derived out of the Gulf of Mexico region [*Carey*, 1958; *MacDonald*, 1967].

We conclude that tectonic rotations may explain anomalous virtual paleomagnetic poles obtained from tectonically disturbed regions and that paleolatitudes calculated from such data may nevertheless be useful in paleogeographic reconstructions.

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